DG INTERN HUB

SOC ANALYST INTERNSHIP

WEEK 2 — TRYHACKME ROOM WRITEUPS

Submitted By: Gunjan

Domain: Cybersecurity / SOC Analyst

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ROOM 5: DNS IN DETAIL

Learning Objective

The objective of this room was to understand how DNS works, learn about different DNS record types, and understand how domain names are translated into IP addresses.

1. Introduction

In this room, I learned about the Domain Name System (DNS) and its role in network communication. DNS helps translate human-readable domain names into IP addresses so that devices can communicate with the correct servers.

For example, when we enter a website name in a browser, DNS helps find the IP address associated with that domain.

2. What Is DNS?

DNS stands for Domain Name System. It works like a directory for domain names and their associated records.

Without DNS, users would generally need to remember IP addresses instead of using easy-to-remember domain names.

3. DNS Hierarchy

DNS uses a hierarchical structure to organise domain names.

- **Root Domain:** The top level of the DNS hierarchy.
- **Top-Level Domain (TLD):** The extension at the end of a domain, such as `.com`, `.org`, or `.uk`.
- **Second-Level Domain:** The main name registered under a TLD.
- **Subdomain:** An additional part of a domain name, such as `blog.example.com`.

A domain name can contain multiple levels, and each level helps identify the domain's position in the DNS hierarchy.

4. DNS Resolution Process

When a user enters a domain name into a browser, the system first checks available cached information. If the required record is not available, a DNS query may be sent to a recursive resolver.

The resolver can query the root server, the relevant TLD server, and the authoritative nameserver to obtain the required DNS record. The result is then returned to the client and may be cached for future use.

5. Common DNS Record Types

- **A Record:** Maps a domain name to an IPv4 address.
- **AAAA Record:** Maps a domain name to an IPv6 address.
- **CNAME Record:** Creates an alias pointing to another domain name.
- **MX Record:** Specifies the mail servers responsible for receiving email for a domain.
- **TXT Record:** Stores text information, often used for domain verification and email security configurations.
- **NS Record:** Identifies the nameservers responsible for a DNS zone.
- **PTR Record:** Used for reverse DNS lookups, mapping an IP address to a domain name.

6. DNS Ports and Protocols

DNS commonly uses port **53**.

- **UDP port 53:** Commonly used for standard DNS queries and responses.
- **TCP port 53:** Used in situations such as zone transfers and responses that require TCP.

DNS traffic is important for network monitoring because many applications rely on it to locate servers.

7. TTL and DNS Caching

TTL stands for Time to Live. It specifies how long a DNS record may remain in a cache before it should be refreshed.

Caching helps reduce repeated DNS queries and improves response times. However, cached records may remain available until their TTL expires.

8. Recursive and Authoritative DNS Servers

Recursive Resolver: Receives DNS queries from clients and obtains the required answer by using its cache or querying other DNS servers.

Authoritative Nameserver: Stores the authoritative DNS records for a domain or DNS zone and provides answers for those records.

Both play important roles in resolving domain names.

9. Practical Learning

During this room, I explored DNS concepts, record types, domain hierarchy, and the resolution process.

I learned how different records provide different types of information. For example, A records provide IPv4 addresses, AAAA records provide IPv6 addresses, and MX records identify mail servers.

10. SOC Analyst Perspective

DNS logs can help SOC analysts investigate suspicious network activity. Unusual domain lookups, repeated queries to unfamiliar domains, or unexpected DNS traffic may require further investigation.

Attackers can also misuse DNS for command-and-control communication or DNS tunnelling. Analysts can review queried domains, source devices, query frequency, and related network events to understand suspicious activity.

11. What I Learned

- I understood the purpose of DNS in network communication.
- I learned about DNS hierarchy and domain names.
- I understood the basic DNS resolution process.
- I learned the purpose of common DNS record types.
- I understood the difference between recursive resolvers and authoritative nameservers.
- I learned why DNS traffic is useful during security investigations.

12. Conclusion

This room helped me understand how DNS translates domain names into useful network information. I also learned that DNS traffic can provide valuable clues during security monitoring and incident investigations.